LOCALE

Ego-Graph Orientation for LLM-Assisted Causal Discovery

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From course presentation to research prototype

Recap: The Landscape We Presented

Three paradigms for LLM-assisted causal discovery:

Paradigm	Example	Context/Query	Queries
Per-edge	MosaCD, chatPC	~200 tokens (one edge)	O(E × K)
Ego-graph	LOCALE (ours)	~500-1000 tokens (neighborhood)	O(N × K)
Whole-graph	CausalFusion	$O(V^2)$ tokens (full DAG)	O(rounds)

MosaCD: shuffled queries + non-collider-first propagation. Best F1 in 9/10 benchmarks.

But each query uses the LLM as a **binary classifier**.
128K context window. ~200 tokens used.

The Gap We Identified

Per-edge (MosaCD): one query = one binary vote. No local consistency.
CI constraints checked after the LLM, not during.

Whole-graph (CausalFusion): entire DAG in context. $O(V^2)$ tokens.
LLM does what Meek rules already do perfectly.

The missing middle: show the LLM one node's neighborhood.
It scores all incident edges **jointly**, sees CI constraints,
reasons about local mechanism consistency.

> The LLM's comparative advantage is contextual reasoning about
> local causal mechanisms — not binary edge classification,
> not global graph editing.

LOCALE: Iterative BFS Expansion

Survey Propagation-style decimation: **compute marginals → fix most confident → simplify → repeat**

Round 1: Query all eligible nodes with ego-graph prompts.
Solve Max-2SAT per node (NCO constraints + LLM soft scores).
Decimate: commit edges where majority vote > 70%.

Round 2+: Query **frontier nodes** — those adjacent to established edges but not yet queried. Prompts now include established orientations as context.
Max-2SAT adds established edges as **hard unary constraints**.
Decimate new high-confidence edges. Repeat.

Convergence: stop when no new edges are decimated, or all edges resolved.

SP Concept	LOCALE Mapping
Variables	Edge orientations $x_e \in \{+1, -1\}$
Factors	Ego-graph queries $f_v(x_{e1},\ldots,x_{ed})$

The Ego-Graph Prompt (Round 2+)

```
CENTER NODE: DrivQuality — "Quality of driving"

NEIGHBORS (3 variables):
  1. DrivingSkill — "Driver's overall skill level"
  2. RiskAversion — "Risk tolerance"
  3. Accident — "Accident severity"

ESTABLISHED ORIENTATIONS (from prior rounds):
  ESTABLISHED (round 1, conf=90%): DrivingSkill → DrivQuality

STATISTICAL CONSTRAINTS:
  R1. DrivingSkill ⊥ Accident | {DrivQuality} — non-collider.
      DrivingSkill and Accident must NOT both point into DrivQuality.

EDGES TO ORIENT:
  - DrivingSkill → DrivQuality  [ESTABLISHED — do not change]
  - DrivQuality -- RiskAversion: orient this
  - DrivQuality -- Accident: orient this
```

Round 1: one query → 4 edges. **Round 2**: context-enriched → 2 remaining edges.

Finding 1: Ego Needs Scale

Ego-graph prompting requires sufficient model capacity.
Same edges, same skeleton, same CI evidence — only model size changes.

Model	Per-Edge	Ego-Graph	Delta
Qwen3.5-**4B**	90.5%	47.4%	**-43.2pp**
Qwen3.5-**9B**	90.5%	80.0%	-10.5pp
Qwen3.5-**27B**	93.7%	94.7%	**+1.1pp**

Joint neighborhood reasoning is a harder task than binary classification.
Small models fail catastrophically. At **27B**, ego matches or wins.

Insurance network, disguised variable names, K=5 votes. XN-001/002/003.

Finding 2: Skeleton Is the Bottleneck

Decomposing directed-edge F1: **skeleton coverage × orientation accuracy**

Network	Skeleton Coverage	Orientation Acc.	Directed F1	MosaCD F1
Insurance	83%	**100%**	88.4%	87%
Asia	88%	**100%**	93.3%	93%
Alarm	88%	89.9%	89.9%	93%
Child	88%	88.0%	88.0%	90%
Hepar2	**52%**	80.6%	58.8%	72%

On edges the skeleton gets right, orientation is **at ceiling**.
The F1 gap vs MosaCD is skeleton coverage, not orientation quality.

Hepar2: 52% skeleton coverage. No orientation method can fix that.

Finding 3: The NCO Discovery

Phase 2 (hard CI constraints) **hurt** on 3/6 networks. Why?

100% of incorrect CI-derived orientation facts are false colliders.

Network	CI Accuracy	False Colliders	False Non-Colliders	P2 Impact
Insurance	92.4%	7	0	+4.2pp
Alarm	90.4%	7	0	+1.3pp
Child	78.9%	12	0	+6.2pp
Sachs	95.5%	1	0	-5.9pp
Asia	90.0%	1	0	-16.7pp
Hepar2	65.1%	52	0	-9.9pp

Non-collider = positive evidence (Z IS in sep set). Always reliable.

Collider = absence of evidence (Z NOT FOUND). Fails with finite samples.

NCO: The Fix and Its Generality

Fix: use only non-collider constraints. 100% reliable.

Network	Ego Only	+ Hard Constraints	+ NCO Only
Insurance	93.1%	97.2%	**98.6%**
Alarm	88.3%	89.6%	89.6%
Sachs	70.6%	64.7%	**70.6%**
Asia	100%	83.3%	**100%**
Hepar2	75.7%	65.8%	**79.3%**
Aggregate	83.7%	81.1%	**86.7%**

NCO never hurts. Hard constraints hurt 3/6 networks.

This is **method-agnostic**: applies to MosaCD, PC, any CI-based method.
Validated at n=1k through n=10k. 100% false-collider rate at n ≥ 5000.

Finding 4: Head-to-Head vs MosaCD

LOCALE uses **open-source Qwen3.5-27B** (local inference).
MosaCD uses **proprietary GPT-4o-mini** (API calls).

Network	LOCALE	MosaCD	Winner
Insurance	**88.4%**	87%	LOCALE (+1.4pp)
Alarm	89.9%	**93%**	MosaCD (-3.1pp)
Child	88.0%	**90%**	MosaCD (-2.0pp)
Asia	**93.3%**	93%	LOCALE (+0.3pp)
Hepar2	58.8%	**72%**	MosaCD (-13.2pp)

Score: 2 wins, 3 losses. Competitive, not dominant.
But: open-source model, local inference, **2.4-4x fewer queries**.
Hepar2 gap is skeleton coverage (52%), not orientation.

Finding 5: Query Efficiency

Ego-graph batching: one query scores **all** incident edges.

Per-edge: one query per edge × K votes = O(E × K) queries
Ego-graph: one query per node × K votes = O(N × K) queries

Across 6 networks: **2.4-4.0x fewer queries** (58-75% savings)

Cost scales with **nodes**, not edges.
As graphs get denser, the savings grow.

At 27B, each ego query costs more tokens but produces
d edges of information instead of 1.
Net result: competitive accuracy at a fraction of the cost.

Sachs (11 nodes, 17 edges): 2.4x. Insurance (27 nodes, 52 edges): 3.8x.

Ablation: What Each Phase Contributes

Full pipeline: **+6.6pp** over per-edge baseline (85.6% → 92.2%)

Phase	What It Does	Cumulative	Delta
PE baseline	Per-edge LLM votes	85.6%	—
+ P1 Ego scoring	Joint neighborhood context	87.0%	+1.4pp
+ P2 NCO constraints	CI-feasible assignments	88.8%	+1.9pp
+ P3 Reconciliation	Dual-endpoint agreement	89.7%	+0.8pp
+ P4 Safety valve	Damage detection + revert	**92.2%**	**+2.5pp**

Safety valve is the largest single contributor.

Not cleanup — core design. Detects when constraints hurt and reverts.

4 networks, 215 edges, orientation accuracy on skeleton edges. XN-012.

What Didn't Work

Dawid-Skene reconciliation (proposed)
Underperforms majority vote. Only 2 annotators per edge — too sparse for EM.

Meek propagation (proposed)
Complete no-op. Max-2SAT already orients everything. Nothing left for Meek.

Skeleton refinement via LLM
0 true positives across 6 networks. Most missing edges are >2 hops apart.
Hepar2 worst: 9 false positives, 0 true positives. Definitively vetoed.

Ego at degree 3 ("valley of confusion")
Too few CI constraints to guide, enough complexity to confuse.
Per-edge wins at d=3. Ego wins at d≥4.

These negatives tell us **where LLMs can't help** in causal discovery.

What MosaCD Got Right, Where LOCALE Adds

MosaCD's insight confirmed: non-collider-first propagation works.
Our NCO finding provides the **theoretical grounding** — it's not just empirically better. We now know why: 100% of CI errors are false colliders.

What LOCALE adds:

1. **NCO as a general principle** — method-agnostic, not MosaCD-specific
2. **F1 decomposition** — skeleton is the binding constraint (reframes the field)
3. **Query efficiency** — 2.4-4x savings via ego-graph batching
4. **Open-source viability** — 27B matches proprietary on most benchmarks
5. **Safety valve** — monotonically non-decreasing pipeline

Where both struggle: skeleton coverage.
Neither can overcome a bad PC skeleton. That's the real frontier.

Limitations

- **Causal sufficiency** assumed — no latent confounders
- **Single model family** — only Qwen3.5 tested (no GPT-4o, Claude, Llama)
- **PC skeleton ceiling** — LOCALE cannot recover edges PC misses
- **n=1000** for most experiments — CI tests degrade on large graphs
- **Calibration** (Venn-Abers) not yet validated — needs held-out graph instances
- **Iterative expansion** not yet tested — current implementation is single-pass

Contributions and What's Next

Five contributions:

1. **NCO discovery**: ~100% of CI errors are false colliders (method-agnostic)
2. **F1 decomposition**: skeleton, not orientation, is the binding constraint
3. **Ego-graph batching**: 2.4-4x query savings
4. **Open-source parity**: 27B matches GPT-4o-mini on 4/5 shared benchmarks
5. **Safety valve**: monotonically non-decreasing pipeline design

What's next:

- Iterative BFS expansion with established context (in progress)
- Better skeletons: hybrid statistical-LLM skeleton learning
- Cross-model validation: GPT-4o, Claude, Llama
- Calibrated selective output (Venn-Abers edge confidence)
- NCO as preprocessing for all CI-based methods

Thank You

Questions?

Code and experiments: projects/locale/ · Qwen3.5-27B on RunPod vLLM